

Báo cáo bài tập EX5: Đánh giá điểm tin dụng sử dụng Decision Tree và Random Forest

Bài 1: Chạy RF baseline

```
#code
RF.fit(X_train, y_train)
y_rf_pred=RF.predict(X_test)
print('Demo RF prediction')
print("-DF acc= {}".format(accuracy_score(y_test, y_rf_pred)))
print("-DF f1  = {}".format(f1_score(y_test,y_rf_pred)))
print("-DF roc auc = {}".format(roc_auc_score(y_test, y_rf_pred)))
print(confusion_matrix(y_test, y_rf_pred))
```

```
Demo RF prediction
-DF acc= 0.81
-DF f1  = 0.875
-DF roc auc = 0.7173939175381656
[[133   8]
 [ 30  29]]
```

Bài 2: đánh giá best_tree

- Prediction performance
- Learning curve

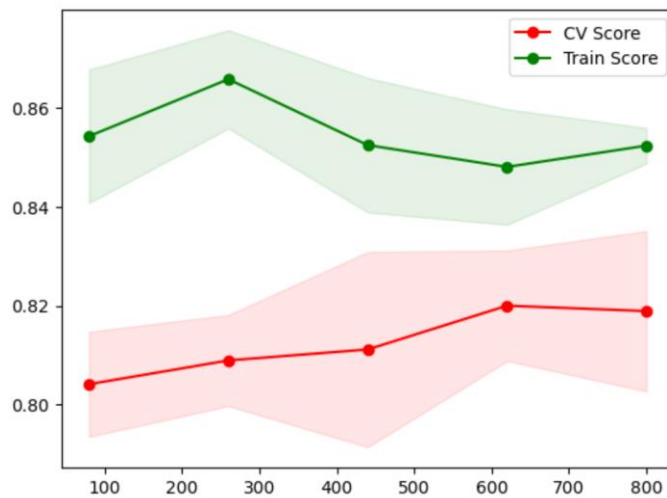
```
In [9]: 
evaluate(best_tree)
plot_learning_curve(best_tree, 'best_tree learning curve', {'train': 'Train Score', 'test': 'CV Score'}, X, y, ylim=None, cv=None,
                    n_jobs=1, train_sizes=np.linspace(.1, 1.0, 5), new_plot=False,
                    idx_color=0)

```

```

Train Accuracy : 0.77125
Train f1 score : 0.8473728106755629
Train roc auc : 0.680523905314024
Train Confusion Matrix:
[[508  51]
 [132 109]]
-----
Test Accuracy : 0.78
Test f1 score : 0.8533333333333334
Test roc auc : 0.6911888448130785
Test Confusion Matrix:
[[128  13]
 [ 31  28]]
```

Out[9]: <module 'matplotlib.pyplot' from 'C:\\Users\\ADMIN\\miniconda3\\Lib\\site-packages\\matplotlib\\pyplot.py'>



Bài 3: thực hành tương tự với RF

```
In [11]: # Bài tập
# Tương tự, thử khảo sát RF với số lượng cây n_estimators
# code
cv_accuracies_by_tree, test_accuracies_by_tree = [], []
num_trees = [5, 10, 15, 20, 30, 50, 75, 100, 150]

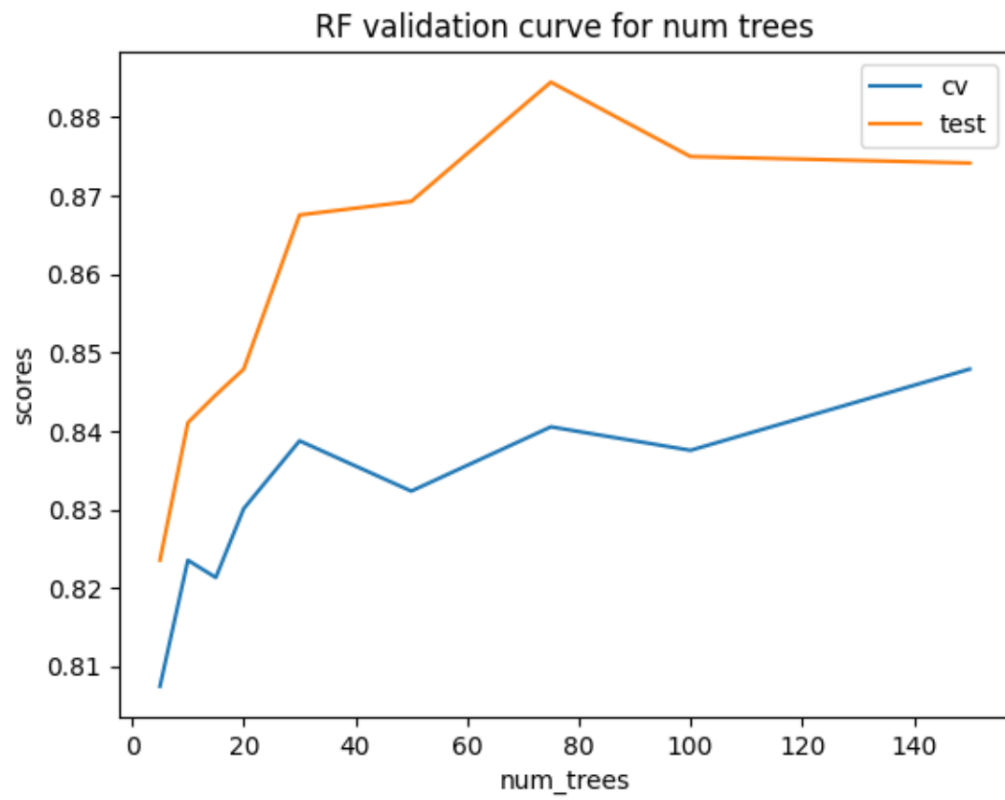
for curr_num_trees in num_trees:
    tree = RandomForestClassifier(random_state=random_state, n_estimators=curr_num_trees)

    val_scores = cross_val_score(estimator=tree, X=X_train, y=y_train, cv=5, scoring='f1')
    cv_accuracies_by_tree.append(val_scores.mean())

    tree.fit(X_train, y_train)
    curr_pred = tree.predict(X_test)
    test_accuracies_by_tree.append(f1_score(curr_pred, y_test))

plt.plot(num_trees, cv_accuracies_by_tree, label='cv')
plt.plot(num_trees, test_accuracies_by_tree, label='test')
plt.legend()
plt.xlabel('num_trees')
plt.ylabel('scores')
plt.title('RF validation curve for num trees')
```

Out[11]: Text(0.5, 1.0, 'RF validation curve for num trees')



```
In [12]: # Sử dụng GridSearch CV
dict_param = {
    'max_depth': [3, 11, 15, 24, 27],
    'min_samples_leaf': [1, 5, 7, 9, 15, 24],
    'max_features': [4, 6, 10, 16],
    'n_estimators': [75]
}
# Bài tập:
# - tìm best_forest
# - Đánh giá best_forest
# - Vẽ learning curve
# - Tìm hiểu feature importance
# Code
best_forest=grid_search('random-forest', n_jobs=-1, dict_param=dict_param)
# - Tìm best_forest
```

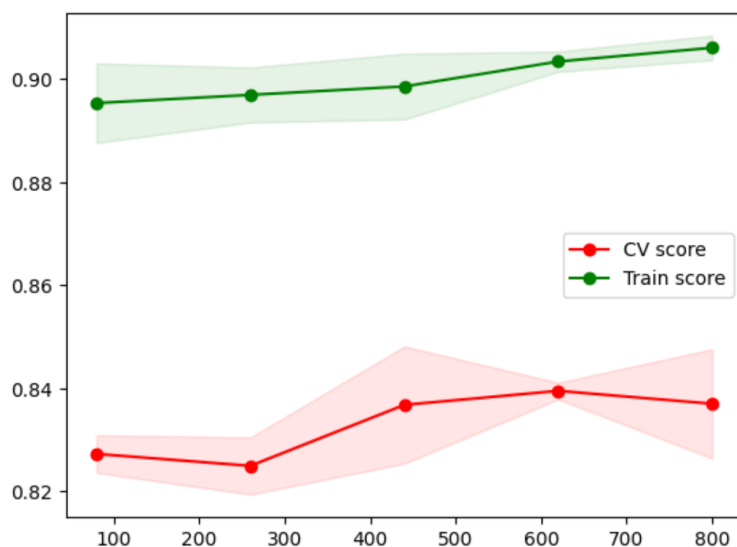
Best modelRandomForestClassifier(max_depth=24, max_features=6, min_samples_leaf=7, n_estimators=75)

```
In [13]: # Code
# - Đánh giá best_forest và vẽ Learning Curve
evaluate(best_forest)
plot_learning_curve(best_forest, 'Learning curve random forest', {'test':'CV score', 'train':'Train score'}, X, y, ylim=None, cv=None)
```

```
Train Accuracy : 0.85125
Train f1 score : 0.9014084507042254
Train roc auc : 0.770815549402831
Train Confusion Matrix:
[[544  15]
 [104 137]]
```

```
-----
Test Accuracy : 0.78
Test f1 score : 0.8589743589743589
Test roc auc : 0.6616179829306407
Test Confusion Matrix:
[[134   7]
 [ 37  22]]
```

Out[13]: <module 'matplotlib.pyplot' from 'C:\\Users\\ADMIN\\miniconda3\\Lib\\site-packages\\matplotlib\\pyplot.py'>



```
In [14]: # Code: Feature importance
importances=best_forest.feature_importances_
indices=np.argsort(importances)[::-1]
print('Top 10 feature importances')
for i in range(10):
    print('Feature ', i, '-', importances[indices[i]])
indices=indices[:10]
plt.figure()
plt.title('Top 10 feature importances')
plt.bar(range(10), importances[indices])
plt.xticks(range(10), indices)
plt.show()
```

```
Top 10 feature importances
Feature  0 - 0.1973859075058331
Feature  1 - 0.12901342850074324
Feature  2 - 0.11528095326814913
Feature  3 - 0.11481067430370316
Feature  4 - 0.09830699764981322
Feature  5 - 0.04735776093900774
Feature  6 - 0.04718497834077803
Feature  7 - 0.04652853541169338
Feature  8 - 0.031052668858855974
Feature  9 - 0.02961875027254437
```

Top 10 feature importances

